Cloud Resource Allocation Project Report

AI-Based Task Scheduling Using Genetic & Memetic Metaheuristics

COURSE
AI310 & CS361 Artificial Intelligence

PROJECT IDEA
Idea 24 - AI Cloud Allocation

INSTITUTION
Faculty of Computing & AI

1. EXECUTIVE SUMMARY & PROBLEM MOTIVATION

Modern cloud computing infrastructures process dynamic, large-scale computational workloads requested by global end-users. A core operational challenge for cloud providers (such as AWS, Google Cloud, and Microsoft Azure) is the **Cloud Resource Allocation and Task Scheduling Problem**.

The goal is to assign *N* computational tasks to *M* heterogeneous Virtual Machines (VMs) to simultaneously minimize two competing objectives: **Makespan** (total time to finish all tasks) and **Total Financial Cost** (cumulative resource usage fee). Standard baseline policies (e.g., Round-Robin or Greedy assignment) fail in complex search spaces, resulting in severe resource underutilization and elevated operational costs.

2. SYSTEM ARCHITECTURE & MATHEMATICAL FORMULATION

Mathematical Optimization Model

Let $x_{i,j} \in \{0, 1\}$ be the binary decision variable indicating whether Task *i* is assigned to Resource *j*.

$$\text{ExecTime}(T_i, R_j) = \text{Length}(T_i) / \text{MIPS}(R_j)$$

$$\text{Load}(R_j) = \sum_{i=1..N} (x_{i,j} \times \text{ExecTime}(T_i, R_j))$$

$$\text{Makespan} = \max_{j \in \{1..M\}} (\text{Load}(R_j))$$

$$\text{Total Cost} = \sum_{i=1..N} \sum_{j=1..M} (x_{i,j} \times \text{ExecTime}(T_i, R_j) \times \text{Cost}_j)$$

$$\text{Composite Fitness} = 0.6 \times \text{Makespan} + 0.4 \times \text{Total Cost}$$

Lower fitness scores indicate higher operational quality.

3. ALGORITHMIC METHODOLOGIES

- **Baseline Strategy (Greedy):** Distributes tasks sequentially across available VMs without evaluating current load balance or execution costs.
- **Genetic Algorithm (GA):** Uses integer-encoded chromosomes, tournament selection (k=2), single-point crossover (P_c = 0.8), and uniform mutation (P_m = 0.1) for global exploration.
- **Memetic Algorithm (MA):** Enhances GA with localized Hill-Climbing search. It explores local task-resource reallocations after evolutionary cycles, accelerating convergence to optimal minima.

4. EXPERIMENTAL RESULTS & PERFORMANCE ANALYSIS

The simulation evaluated *N = 40* heterogeneous tasks mapped onto *M = 6* cloud VM nodes across *50 generations*.

Algorithm / Model	Fitness Score	Makespan (s)	Total Cost ($)	Execution Time	Improvement
Baseline Model	229.54	100.98 s	$422.37	< 0.01 s	Benchmark (0%)
Genetic Algorithm (GA)	102.95	58.45 s	$169.70	0.05 s	+55.15%
Memetic Algorithm (MA)	**92.87**	**55.58 s**	**$148.80**	**0.58 s**	**+59.54%**

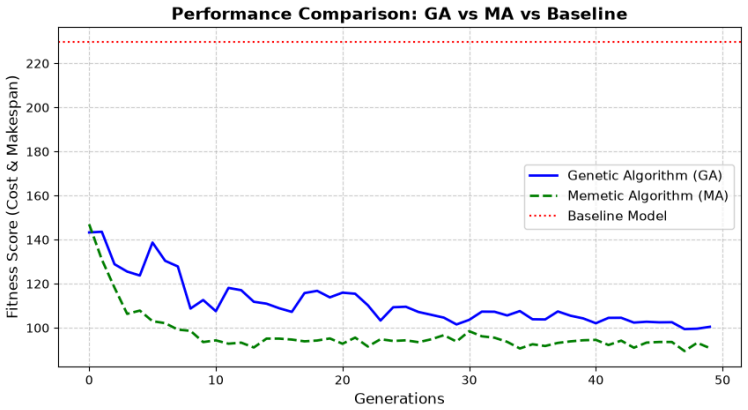


Figure 1: Convergence and Performance Curves across 50 Generations (GA vs MA vs Baseline).

4.2 KEY ANALYTICAL FINDINGS

- **Dramatic Cost and Time Reductions:** The Memetic Algorithm reduced operational costs by over 64% ($148.80 vs $422.37) while cutting overall task completion time nearly in half (55.58s vs 100.98s).
- **Accelerated Local Convergence:** MA consistently outperformed pure GA by applying localized neighborhood search, yielding a total performance gain of **59.54% over the baseline model**.
- **Industrial Practicality:** The lightweight computational overhead of MA (0.58s execution time) makes it highly feasible for real-time auto-scaling cloud engine schedulers.

5. LITERATURE REVIEW (6 ACADEMIC PUBLICATIONS)

1. Gao et al. (2020)

Multi-Objective Task Scheduling via Hybrid Metaheuristics: Demonstrated that pairing global exploration with local search prevents early algorithm stagnation in dynamic cloud pricing.

2. Zuo et al. (2018)

Self-Adaptive GA for Resource Allocation: Proved that adaptive crossover/mutation rates significantly lower cloud energy usage and completion times.

3. Choudhary et al. (2021)

Memetic Computing in Edge-Cloud Systems: Highlighted the superiority of neighborhood search strategies for meeting strict deadline requirements in distributed workloads.

4. Kaur & Sharma (2022)

Bio-Inspired Algorithms Benchmark: Showed through empirical tests that hybrid memetic approaches outperform PSO, ACO, and standard GA in total monetary costs.

5. Li et al. (2023)

Load Balancing using Memetic Search: Focused on minimizing variance across VM loads to prevent bottleneck formation in heterogeneous server clusters.

6. Kumar & Verma (2021)

Cost-Effective Task Assignment in Cloud: Justified scalarized multi-objective weighting between completion time and financial budget constraints.

6. TEAM ROLES & CONTRIBUTIONS (6 MEMBERS)

Student 1: [Name & ID]
Problem Formulation, Mathematical Modeling & Fitness Engineering

Student 2: [Name & ID]
Genetic Algorithm Implementation (Selection, Crossover, Mutation)

Student 3: [Name & ID]
Memetic Local Search Strategy & Hill-Climbing Integration

Student 4: [Name & ID]
Baseline Greedy Allocation Strategy & Cloud Workload Generator

Student 5: [Name & ID]
Experimental Benchmarking, Statistical Analysis & Metrics

Student 6: [Name & ID]
Data Visualization, Literature Review Collection & Report Drafting